MulTCIM: Digital Computing-in-Memory-Based Multimodal Transformer Accelerator With Attention-Token-Bit Hybrid Sparsity

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***Abstract*— Multimodal Transformers are emerging artificial intelligence (AI) models that comprehend a mixture of signals from different modalities like vision, natural language, and speech. The attention mechanism and massive matrix multiplications (MMs) cause high latency and energy. Prior work has shown that a digital computing-in-memory (CIM) network can be an efficient architecture to process Transformers while maintaining high accuracy. To further improve energy efficiency, attention-token-bit hybrid sparsity in multimodal Transformers can be exploited. The hybrid sparsity significantly reduces computation, but the irregularity also harms CIM utilization. To fully utilize the attention-token-bit hybrid sparsity of multimodal Transformers, we design a digital CIM-based accelerator called MulTCIM with three corresponding features: The long reuse elimination dynamically reshapes the attention pattern to improve CIM utilization. The runtime token pruner (RTP) removes insignificant tokens, and the modal-adaptive CIM network (MACN) exploits symmetric modal overlapping to reduce CIM idleness. The effective bitwidth-balanced CIM (EBB-CIM) macro balances input bits across in-memory multiply-accumulations (MACs) to reduce computation time. The fabricated MulTCIM consumes only 2.24 μJ/Token for the ViLBERT-base model, achieving 2.50×–5.91× lower energy than previous Transformer accelerators and digital CIM accelerators.**

***Index Terms*— Computing-in-memory (CIM), dataflow, hybrid sparsity, multimodal transformers, reconfigurable architecture.**

I. Introduction

ARTIFICIAL intelligence (AI) has one important goal of approaching human's multimodal perception, which is able to comprehend a mixture of vision, natural language, speech, and so on. In recent algorithm advances [12], [14], [16], [17], multimodal Transformers are promising models that can learn from different modalities as human perception, achieving excellent results on multimodal AI tasks such as video question answering, multilingual image retrieval, and action prediction.

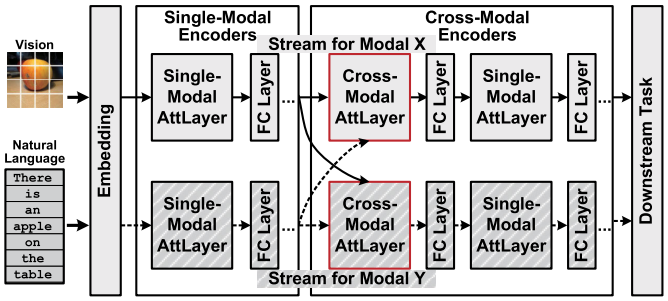


Fig. 1. Typical multimodal transformer with cross-modal attention.

In vanilla Transformers, the attention mechanism and large numbers of matrix multiplications (MMs) cause high latency and energy, demanding specialized hardware for efficient inference. For example, Wang et al. [22] propose a digital Transformer accelerator with approximate computing techniques to improve energy efficiency. Since digital computing-in-memory (CIM) has proven to be an efficient MM computation architecture with high precision support [4], [6], TranCIM constructs its architecture based on digital CIM with pipeline/parallel reconfigurable modes to further improve efficiency [20]. For attention layers, the architecture works in the pipeline mode to generate Matrix Q, K, V and dynamically consumes them without off-chip memory access. For fully connected (FC) layers, all the CIMs store the weights and work in parallel.

Multimodal Transformers also have much MM computation amount and introduce a cross-modal attention mechanism to vanilla Transformers for learning joint information from different modalities, as shown in Fig. 1. The new model structure offers attention-token-bit hybrid sparsity opportunities. Previous Transformer accelerators usually focus on reducing computation in the attention mechanism [7], [8], [13], [15], [20], [22], but fully exploiting the hybrid sparsity can bring higher end-to-end efficiency. In this article, we will discuss how to overcome the challenges of utilizing hybrid sparsity on the promising digital CIM network architecture proposed in TranCIM [20]. A simplified CIM network architecture is shown on the top-left of Fig. 2. It is

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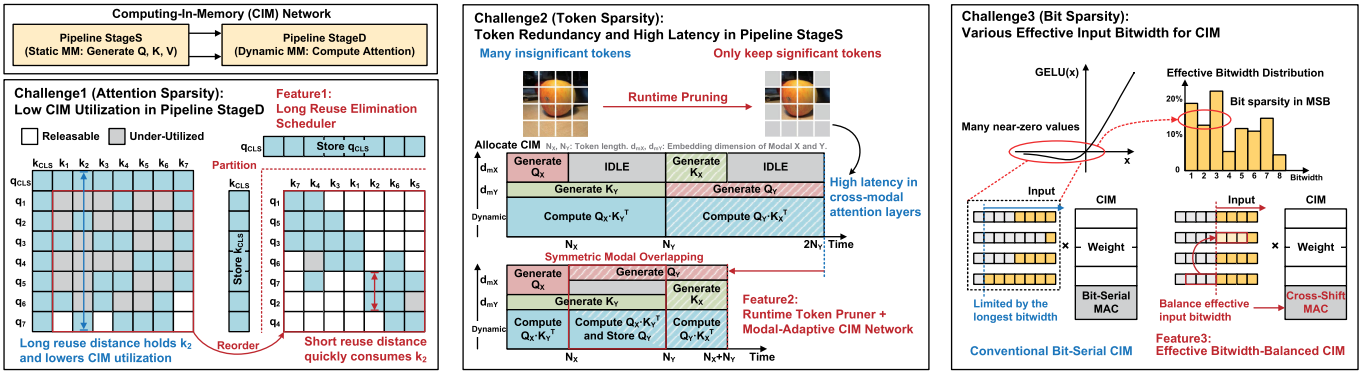


Fig. 2. Multimodal transformer's hybrid sparsity raises challenges for CIM accelerator design. MulTCIM has corresponding features that target the challenges.

configured as a two-stage pipeline for attention layers: Pipeline StageS is for Static MM of Q, K, V generation, and Pipeline StageD is for Dynamic MM of attention computation. Detailed challenge analysis is illustrated in the rest part of Fig. 2.

Challenge 1 (Attention Sparsity): The attention matrix has irregular sparse patterns that may lead to long reuse distance (like k_2 in the example). For instance, the distance can cover 78.6%–81.7% of the token count in the ViLBERT-base model. Thus, in Pipeline StageD, the CIM macros have to hold infrequently used weights for a long time, leading to low CIM utilization.

Challenge 2 (Token Sparsity): Multimodal signals such as images, video, and audio have diverse significance in tokens, so we can use token pruning to reduce computation. As studied in [23], pruning the redundancy in image tokens can lead to over 1.6× speedup with negligible accuracy loss. However, different modalities have diverse token length. For cross-modal attention layers, generating Q, K for different modalities causes CIM idleness and high latency in Pipeline StageS.

Challenge 3 (Bit Sparsity): Activation functions like Softmax and GELU produce many near-zero values that expose bit sparsity in the most-significant bits (MSBs). The bit sparsity can be 58.03% in Softmax's MSB and 54.60% in GELU's MSB. Besides, as token sparsity reduces value locality, a CIM macro has more variance in effective bitwidth (EB) for the same group of inputs. Conventional CIM's bit-serial multiply-accumulation (MAC) scheme makes computation time limited by the longest bitwidth.

This article proposes a multimodal Transformer CIM accelerator, called MulTCIM, with attention-token-bit hybrid sparsity exploitation. MulTCIM uses digital CIM macros as the computing units of the accelerator and has the following features corresponding to each level of sparsity.

1) For **Challenge 1**, we implement a long reuse elimination scheduler (LRES) to dynamically reshape the attention matrix into a pattern that is similar to regular global+local sparsity, unlike the naive attention computation that generates Q, K, V tokens in order. Thus, weights stored in CIM can be reused more frequently to improve CIM utilization in Pipeline StageD.
2) For **Challenge 2**, we design a runtime token pruner (RTP) to remove insignificant tokens and a modal-adaptive CIM network (MACN) to dynamically divide all CIM cores into Pipeline StageS and StageD. In cross-modal attention layers, MACN exploits symmetric modal overlapping to mix two streams of attention, thus reducing CIM idleness and Q, K generation latency in Pipeline StageS.
3) For **Challenge 3**, we use an EB-balanced CIM (EBB-CIM) macro architecture to balance input bits across in-memory MACs by performing EB detection and bit equalization, thus reducing the computation time.

This article is the journal extension version of our ISSCC'23 work [21], providing higher-level insights to readers with many additional materials. One of the most important contributions is handling the irregular dataflow caused by hybrid sparsity with inter- and intra-CIM architecture optimizations. LRES, RTP, and MACN are inter-CIM designs that optimize how to organize CIM macros. EBB-CIM is an intra-CIM design that optimize the CIM macro's inner structure. The rest of this article is organized as follows: Section II provides background and motivation. Section III proposes MulTCIM's overall architecture. The three architecture features are described in Sections IV–VI, respectively. Section VII presents our experimental results on the 28-nm MulTCIM chip. Section VIII concludes this article.

II. Background and Motivation

A. Multimodal Transformers

Fig. 1 shows a typical multimodal Transformer model with a stacking structure of single-modal and cross-modal encoders. To process two modalities, the encoders can be divided into two streams for Modal X and Modal Y, respectively. In the example, X is for vision (upper stream), and Y is for language (lower stream). The single-modal attention layers are similar to those in vanilla Transformers. Cross-modal attention layers are introduced to enable information exchange between modalities. Fig. 3 illustrates the one in the stream of Modal X: The queries come from Modal X ($Q_X = X \cdot W_Q$), while the keys and values are from Modal Y ($K_Y = Y \cdot W_K$, $V_Y = Y \cdot W_V$). Similarly, cross-modal attention layers in the stream of Modal Y generate queries from Modal Y with keys and values from Modal X. The rest attention computation is similar to the conventional attention mechanism. Consequently, cross-modal attention produces features for one

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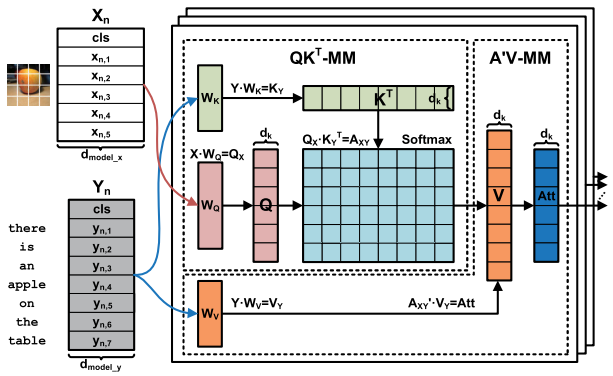


Fig. 3. Cross-modal attention layer in the stream of Modal X.

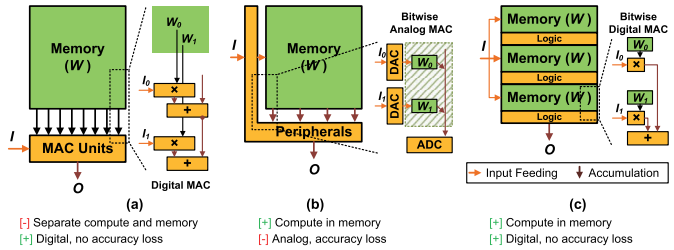


Fig. 4. Overview of different AI accelerator architectures. (a) Digital architecture. (b) Analog CIM. (c) Digital CIM.

modality conditioned on the other, such as vision-conditioned language attention and language-conditioned vision attention. The joint information enables multimodal Transformers to comprehend deeper knowledge behind different modalities.

B. Digital CIM

AI accelerators have been studied for a long time. Many of them are based on digital architectures with separate MAC units and memory [see Fig. 4(a)] [1], [3], [10], [11], [19], [22], [25]. However, as AI model size and computation amount increase, the massive data movements between MAC units and memory affect the overall energy efficiency. The problem is more severe for large-scale Transformer models.

Digital CIM is a promising architecture that achieves high efficiency and high accuracy by integrating bitwise digital MAC to SRAM [4], [6], [18], as shown in Fig. 4(c). For one neural network layer represented by $O = I \times W$, the digital architecture should load the weight matrix (W) from memory to MAC units row by row, causing massive data movements that reduce energy efficiency. Compared with the conventional architecture, digital CIM directly stores the weight matrix in memory and performs MAC operations locally without reading weights out. Moreover, digital CIM can reuse SRAM's inner logic to construct bitwise multipliers to save more transistors for lower energy consumption. Since digital CIM performs computation in the digital domain, it also has no accuracy as the conventional digital architecture. Compared with analog CIM [9], [24], [26] [see Fig. 4(b)], digital CIM does not need to convert signals between analog and digital domains, avoiding nonideal issues caused by signal margin and process variation. Therefore, digital CIM can easily support precision higher than 8 b, which is important to maintain high accuracy for emerging AI models such as Transformers.

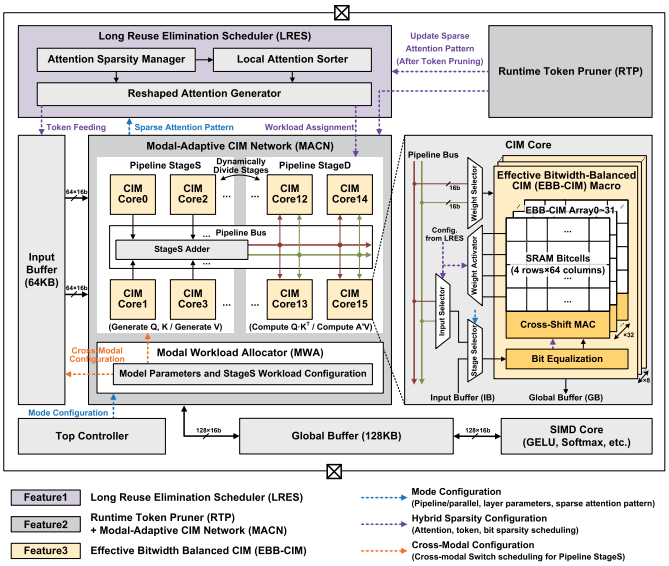


Fig. 5. MulTCIM accelerator's overall architecture.

TranCIM [20] is a representative work for digital CIM-based Transformer accelerators. Considering Transformer's model structure, TranCIM focuses on optimizing inter-CIM network architecture design and utilizes pipeline/parallel modes for different Transformer layers. Meanwhile, TranCIM exploits pretrained regular attention sparsity, which is widely used in vanilla Transformers [2], [5], [27], but not common in multimodal Transformers. This is because the sparse attention pattern reflects the correlation between different tokens. In vanilla (single-modal) Transformers, a token tends to be more correlated with the tokens in the neighborhood or at fixed positions, which leads to more regular attention sparsity [5]. However, in multimodal Transformers, the sequence length and meaning of tokens are various for different modalities. The correlation between different tokens is not limited to certain positions, so it is not easy to regularize the attention sparsity with a fixed pattern. Due to the more irregular attention sparsity of multimodal Transformers, dynamic dataflow scheduling should be carefully considered in accelerator architecture design.

C. Motivation of MulTCIM

Besides attention sparsity, token and bit sparsity can be utilized to improve energy efficiency. Multimodal Transformers offer not only hybrid sparsity opportunities, but also raise new challenges when we exploit them. As summarized in Section I, the challenges mainly come from irregularity in the three levels of sparsity, leading to lower CIM utilization that harms energy efficiency. In this article, we will discuss how to overcome the challenges on the basis of the promising digital CIM network architecture proposed by TranCIM [20]. Both inter- and intra-CIM innovations are studied to enable flexible utilization of the available CIM resources.

III. MulTCIM Accelerator Architecture

The MulTCIM accelerator's overall architecture is shown in Fig. 5, with three features highlighted in different colors.

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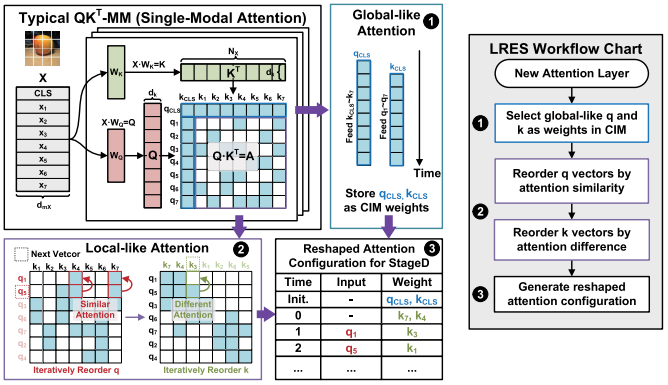


Fig. 6. Long reuse elimination workflow, with a QK^T-MM example.

It comprises a MACN with 16 CIM cores, an RTP, an LRES, a 64-kB input buffer, a 128-kB global buffer, an SIMD core, and a top controller. The CIM cores are connected through MACN's pipeline bus, and each core has eight EBB-CIM macros. The dashed arrows indicate three types of configurations in MulTCIM: Mode configuration for pipeline/parallel modes, layer parameters, and a sparse attention pattern; Hybrid sparsity configuration for attention, token, bit sparsity-related dynamic scheduling; Cross-modal configuration for modality switch scheduling in Pipeline StageS.

MulTCIM's computing resources are organized in a digital CIM network with reconfigurable modes for different layers in a multimodal Transformer. For FC layers, MulTCIM works in the parallel mode with all CIM cores storing the current layer weights and producing outputs. For attention layers (QK^T-MM and $A'V$-MM), CIM cores work in the pipeline for less off-chip access of intermediate data [20]. The first pipeline stage is StageS for Static MM of Q, K, V generation, and the second pipeline stage is StageD for Dynamic MM of attention computation. Take QK^T-MM, for example. RTP first prunes insignificant input tokens. LRES then produces configurations of sparse token and attention scheduling for the input buffer and MACN's StageD. The CIM cores in StageS store weight matrices (W_Q, W_K) and load inputs from the input buffer to generate matrixes Q and K. The output q, k vectors are merged in the StageS adder and sent to StageD through the pipeline bus. LRES controls the StageS output's destination core in StageD for weight writing or input feeding, so StageD can compute the attention matrix $A = Q \cdot K^T$. Specifically for cross-modal attention layers, MACN utilizes the modal workload allocator (MWA) to adjust the StageS workload when switching streams from Modal X to Modal Y. MACN's outputs are finally stored in the global buffer with activation functions performed in the SIMD core.

IV. LRES for Attention Sparsity Optimization

A. Long Reuse Elimination Workflow

Fig. 6 illustrates the LRES workflow. As we have discussed in **Challenge 1**, the irregular sparse pattern of the attention matrix A causes long reuse distance for the weights stored in Pipeline StageD's CIM. We use a QK^T-MM example (top-left of Fig. 6) to clarify the long reuse issue. In the example, if k_2 is stored in CIM, it should be held for a long time to read q vectors from q_{CLS} to q_7, but the CIM only works when q_{CLS}, q_2, q_7 vectors are fed as inputs. To improve CIM utilization in Pipeline StageD, we design LRES to dynamically reshape the attention matrix into a pattern similar to global + local sparsity used in prior sparse Transformer algorithms [2], [27]. After reshaping, the attention matrix can be partitioned: The global-like attention's weight vectors are stored in CIM for longer time. The local-like attention's weight vectors are more frequently consumed and updated in CIM to eliminate unnecessary long reuse distance. The suffix "-like" means that the reshaped attention is not the perfect global + local pattern defined in [2] and [27]. The solution works for both single-modal and cross-modal attention.

The long reuse elimination workflow has three steps. ❶ For each new attention layer, we first select global-like attention's q, k vectors. In the example, q_{CLS} and k_{CLS} are selected, because they have attention to many vectors (k_{CLS}–k_7 and q_1–q_7, respectively). They will be stored in StageD's CIM for longer time and keep high CIM utilization during computation. ❷ For the rest attention matrix, q, k vectors are reordered to form a local-like pattern with k as weights and q as inputs in StageD's CIM. Local-like means k vectors are frequently consumed and dynamically replaced by new k vectors generated by StageS, so there is less chance for CIM idleness. ❸ Reshaped attention configurations are generated to indicate the sequence for input feeding and weight writing in Pipeline StageD's CIM.

B. LRES Architecture

After discussing the long reuse elimination workflow, we introduce the LRES architecture and explain how the proposed workflow is implemented in hardware, as illustrated in Fig. 7. LRES has three components: an attention sparsity manager, a local attention sorter, and a reshaped attention generator, which are corresponding to the workflow's three steps for global-like attention, local-like attention, and configuration generation. The attention sparsity manager stores the initial sparse attention pattern. It updates the attention pattern based on the runtime token pruning information from RTP. Then the manager iteratively computes row-wise and column-wise sums of the sparse attention pattern. If the sums are larger than a threshold, the corresponding q, k vectors are selected for global-like attention. Their indices are stored in the global attention FIFO.

The local attention sorter reorders the rest attention as a local-like pattern. The similarity-based q-sorter row-wisely reorders q and determines the input feeding sequence. Then, the difference-based k-sorter column-wisely reorders k and determines the weight writing sequence. In this example of Fig. 6, q_5 has the greatest similarity to q_1, so it is selected as the next input q vector and the k vectors already in CIM can be reused continually. We compare the difference between the k vectors with attention to q_5 and q_1 and decide that k_3 should be the next vector stored in CIM. The vector indices for input feeding and weight writing are stored in the local-q and local-k FIFOs.

LRES's outputs are configurations for the CIM macros under the reshaped attention sparsity. They are produced by

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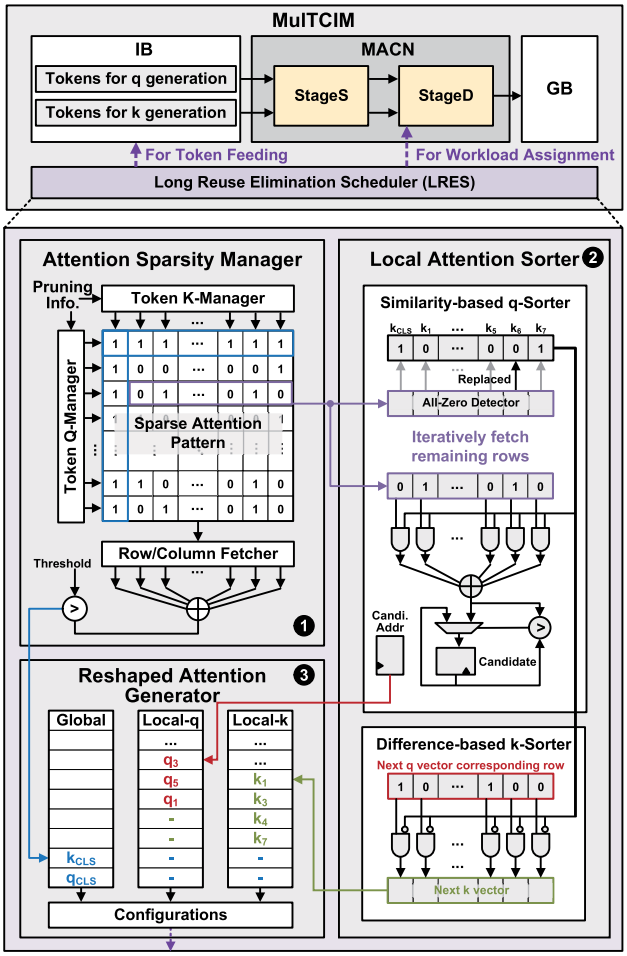


Fig. 7. LRES architecture and the attention sparsity configuration flow.

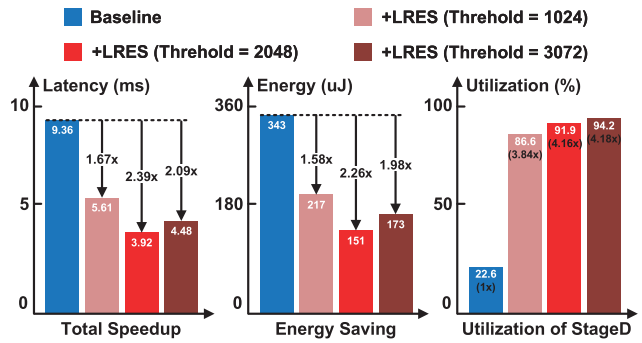


Fig. 8. Evaluation for the LRES architecture. ViLBERT-base QK^T-MM parameters: INT16, $N = 4096$, $d_m = 512$, $d_k = 32$, and attention sparsity = 70%. Baseline: MulTCIM computes Q, K, attention in order (without LRES).

the reshaped attention generator based on the sequence in the three FIFOs for global, local-q, and local-k attention. The configurations consist of two parts: The first part is to control the input buffer to feed tokens for Q, K generation in StageS. The second part is to control each CIM macro's workload assignment in StageD.

C. Technique Evaluation

Fig. 8 shows the evaluation for the LRES architecture. The benchmark is a typical QK^T-MM in ViLBERT-base

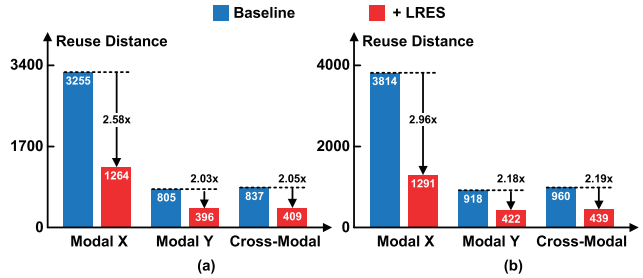


Fig. 9. Average reuse distance of the attention matrices in (a) ViLBERT-base model and (b) ViLBERT-large model. Token count $N_X = 4096$ and $N_Y = 1024$.

with parameters of INT16 precision, $N = 4096$, $d_m = 512$, $d_k = 32$, and attention sparsity = 70%. The baseline (blue bar) disables MulTCIM's LRES to compute Q, K, and attention in order. The proposed LRES dynamically reshapes the attention matrix as a global + local-like sparse pattern. The bars in different red scales represent LRES with different thresholds (1024, 2048, 3072) for selecting global-like and local-like attention. A larger threshold means fewer q, k vectors are selected for global-like attention. Meanwhile, more vectors are selected for local-like attention, so the weights in StageD's CIM can be reused more frequently to improve CIM utilization. However, if the threshold is too large (such as 3072), the increasing local attention vectors cost much longer time and higher energy for LRES's sorters to reshape the attention. Therefore, we choose the medium threshold (2048) at the sweet point, which achieves 2.39× speedup, 2.26× energy saving over the conventional in-order computing approach, and 4.16× higher CIM utilization for Pipeline StageD.

Fig. 9 provides a deeper look into the reuse distance before and after long reuse elimination on the ViLBERT-base and ViLBERT-large models. We separately analyze the average reuse distance of their attention matrices in single-modal attention layers and cross-modal attention layers. In ViLBERT-base, the original reuse distance is 3255 for Modal X, 805 for Modal Y, and 837 for cross-modal attention. 3255 is close to Modal X's token count ($N_X = 4096$). 805 and 837 are close to Modal Y's token count ($N_Y = 1024$). This means StageD's CIM should hold the weights almost throughout the attention computation. After long reuse elimination, the reuse distance decreases significantly, 1264 for Modal X, 396 for Modal Y, and 409 for cross-modal attention. ViLBERT-large presents a similar trend as ViLBERT-base with reuse distance reduction of 2.18×–2.96×.

V. RTP and MACN for Token Sparsity Optimization

A. Token Pruning Based on Class (CLS) Scores

Previous efficient Transformer study has demonstrated that the CLS token's attention to other normal tokens can characterize their significance, so it can be used as the reference for token pruning in both single-modal and cross-modal attention layers [23]. Fig. 10 demonstrates the cross-modal attention mechanism with token pruning. In Layer$_{n-1}$, Modal X and Y's CLS scores derive from their attention matrix's first row. The smaller values in the scores indicate that the corresponding tokens in Layer$_n$ can be pruned.

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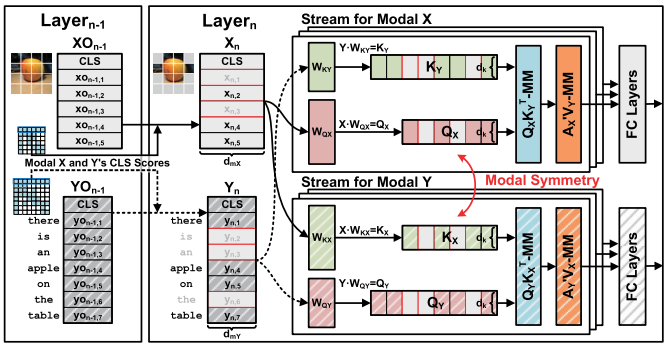


Fig. 10. Cross-modal attention mechanism with token pruning.

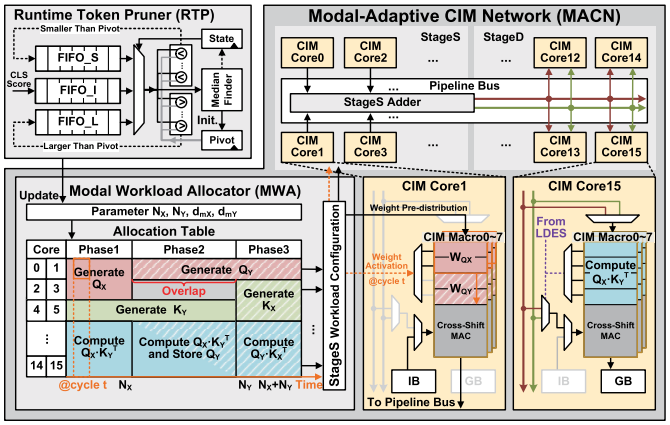


Fig. 11. RTP and MACN architecture.

Different modalities have diverse token length in the cross-modal attention layer, and token pruning increases the diversity. As shown in Fig. 10, since Modal X has fewer tokens than Modal Y, generating Q_X will cost shorter time than generating K_Y. The CIM macros for Q_X generation suffer from idleness when waiting for the new workload, leading to high latency in Pipeline StageS (**Challenge 2**).

B. RTP and MACN Architecture

We will discuss the RTP and MACN architecture with a special focus on how to handle **Challenge 2** in cross-modal attention layers. As illustrated in Fig. 11, RTP receives the previous layer's CLS scores and selects the current layer's top-n most-significant tokens. Based on the median-of-medians top-n algorithm, RTP reduces selection time complexity from $O(n^2)$ to $O(n)$, thus hiding the latency under the previous layer computation. MACN comprises an MWA, 16 CIM cores, and a pipeline bus. For FC layers, all the CIM cores work in the parallel mode. For attention layers, MWA divides CIM cores into Pipeline StageS and StageD based on the original layer parameters and RTP's pruning information. The StageS workload is configured by MWA according to an allocation table, while the StageD workload is scheduled by LRES.

Specifically for **Challenge 2** in cross-modal attention layers, MWA exploits symmetric modal overlapping to mix the processing of two streams. The conventional layer-by-layer method processes the stream for Modal X and then switches to the stream for Modal Y. In our method, Q, K

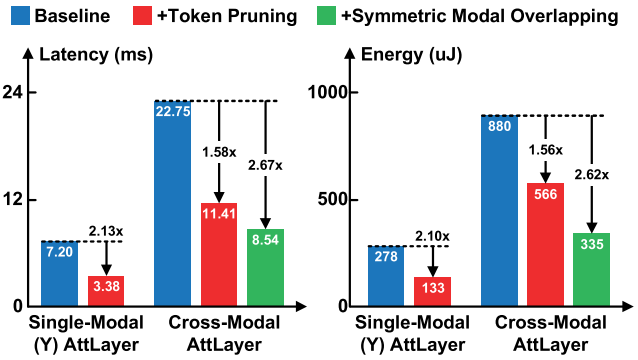


Fig. 12. Evaluation for the RTP and MACN architecture, on ViLBERT-base's attention layers: For Modal X, INT16, $N_X = 4096$, $d_{mX} = 512$, $d_k = 32$, and token sparsity = 30%. For Modal Y, INT16, $N_Y = 1024$, $d_{mY} = 768$, $d_k = 32$, and token sparsity = 50%. Baseline: MulTCIM without token pruning and symmetric modal overlapping.

generation of two different streams are overlapped to reduce CIM idleness. In the example shown in Fig. 11, the allocation table indicates that CIM Core0–5 are configured for StageS, and CIM Core6–15 are configured for StageD. Due to the CIM macro's partial activation structure, a macro can store weights from different modalities and activates one modality at a time. For instance, Core0 and 1 store both W_{QX} and W_{QY} in their macros, so they can first generate Q_X and then generate Q_Y without switching latency. We divide the computation into three phases.

1) *Phase1 for Modal X's Stream:* Core0–3 generate Q_X, and Core4 and 5 generate K_Y. The rest cores compute $Q_X \cdot K_Y^T$. Fig. 11 demonstrates the status of Core1 and Core15 at cycle t of Phase1.
2) *Phase2 for Symmetric Modal Overlapping:* Since Modal X's token count N_X is smaller than Modal Y's token count N_Y, Q_X generation finishes earlier than K_Y generation. Core0 and 1 switch to Modal Y's stream and use W_{QY} to generate Q_Y.
3) *Phase3 for Modal Y's Stream:* Core0 and 1 generate Q_Y, and Core2–5 generate K_X. The rest cores compute $Q_Y \cdot K_X^T$.

Since cross-modal attention exchanges the two modalities' keys to compute QK^T-MM, there exists modal symmetry between the two streams. As a result, Q_X, K_X generation consumes the same CIM core count with the same latency, while Q_Y, K_Y generation also consumes the same CIM core count with the same latency. The modal symmetry helps achieve better overlapping with fewer idle CIM resources.

C. Technique Evaluation

Fig. 12 shows the latency and energy evaluation for the RTP and MACN architectures. The benchmarks are single-modal and cross-modal attention layers in the ViLBERT-base model. Detailed benchmark parameters are as follows: For Modal X, INT16, $N_X = 4096$, $d_{mX} = 512$, $d_k = 32$, and token sparsity = 30%. For Modal Y, INT16, $N_Y = 1024$, $d_{mY} = 768$, $d_k = 32$, and token sparsity = 50%. The baseline (blue bar) is realized on MulTCIM by disabling token pruning and symmetric modal overlapping. The single-modal

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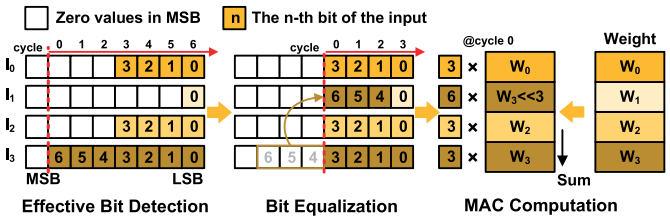


Fig. 13. EB-balanced multiplication.

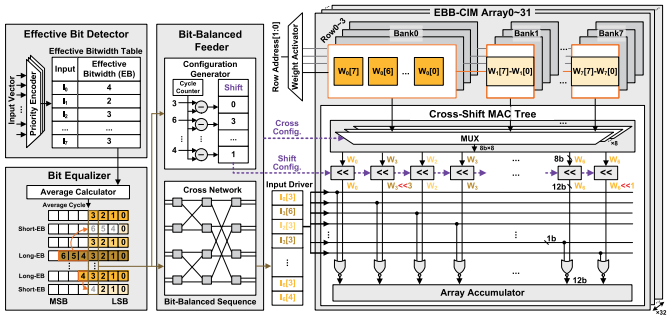


Fig. 14. EBB-CIM macro architecture.

attention layer achieves reductions of 2.13× in latency and 2.10× in energy, because insignificant tokens are pruned by RTP. The cross-modal attention layer obtains further benefits by MACN's symmetric modal overlapping. There is less CIM idleness after MACN mixes the processing of two streams. The total reductions reach 2.67× in latency and 2.62× in energy.

VI. EBB-CIM for Bit Sparsity Optimization

A. EB-Balanced Multiplication

Fig. 13 uses the multiplication between a four-element input vector (I_0–I_3) and weight vector (W_0–W_3) to explain the EB-balanced multiplication for our CIM design. Since the data precision is INT8, the conventional bit-serial computation requires eight cycles for computation. We observe that activation functions like GELU and Softmax produce many near-zero values that expose bit sparsity in MSB. The EB of I_0–I_3 becomes 4, 1, 4, and 7, if the zeros in MSB are removed. Since the computation time is limited by the longest EB (I_3), the cycle count is only reduced to 7. Then, the long-EB element's bits are assigned to the short-EB element's MSB position, producing a bit-balanced input sequence. By assigning I_3's fourth–sixth bits to I_1, the bit count is all the element positions are balanced, which can be sent to the weight vector in four cycles. To make use of the bit-balanced input sequence, the weight elements should be moved to the right position in each cycle to perform correct multiplication with the balanced input bits. Cycle 0 computes $I_0[3] \times W_0 + I_3[6] \times (W_3 \ll 3) + I_2[3] \times W_2 + I_3[3] \times W_3$. Similarly, W_3 keeps staying in the W_1 position at Cycles 1 and 2. Cycle 3 computes the normal MAC operation without moving weights ($I_0[0] \times W_0 + I_1[0] \times W_1 + I_2[0] \times W_2 + I_3[0] \times W_3$).

B. EBB-CIM Architecture

Fig. 14 shows the EBB-CIM macro architecture. An EBB-CIM macro comprises 32 EBB-CIM arrays that share the same inputs, an effective bit detector, a bit equalizer, and a bit-balanced feeder. Each EBB-CIM array has four rows × 64 columns 6T-SRAM bitcells (separated to eight banks) and a cross-shift MAC tree. We exploit a full-digital CIM architecture with 4:1 activation similar to [6]. The four rows reused the same CIM logic in a time-sharing style. The EB detector receives an eight-element input vector and detects each element's EB with priority encoders. The bit equalizer compares each element's EB with the average EB and classifies the elements. Those elements with EB longer than the average are marked as "long-EB," while those with EB shorter than the average are marked as "short-EB." If a long-EB and short-EB pair is found in neighboring elements, the long-EB element's bits will be assigned to the short-EB element for bit equalization. The bit-balanced feeder fetches the input bit sequence from MSB to LSB and generates cross and shift configurations for the in-memory MAC tree. The cross-configuration indicates where to move the weight, and the shift configuration indicates the number of bits for left-shifting. A weight vector is stored across the banks in the same row and read out simultaneously. In the INT8 example, the CIM cross-shift MAC tree crosses and shifts eight weights to multiply the eight input bits from the bit-balanced feeder at the correct position.

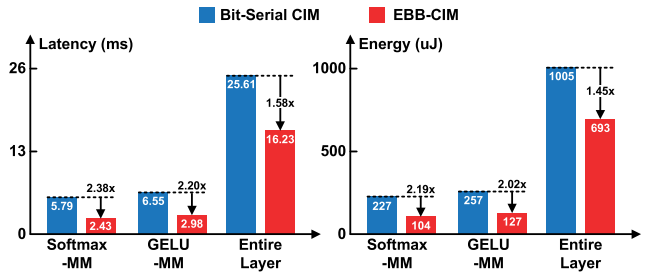


Fig. 15. Evaluation for the EBB-CIM architecture on ViLBERT-base's single-modal layer. Attention parameters: $N = 4096$, INT16, $d_m = 512$, and $d_k = 32$. FC parameters: $N = 4096$, INT8, $d_{\text{ff}} = 4$, and $d_m = 2048$. Softmax, GELU-MM: Results from Softmax or GELU multiplied by weights.

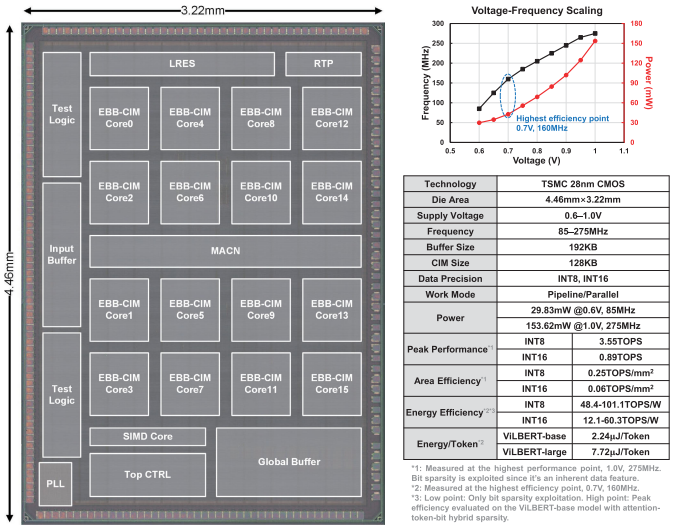
C. Technique Evaluation

Fig. 15 shows the evaluation for the EBB-CIM architecture. The benchmark is one entire single-modal layer stack in the ViLBERT-base model. For attention layers, $N = 4096$, INT16, $d_m = 512$, and $d_k = 32$. For FC layers, $N = 4096$, INT8, $d_{\text{ff}} = 4$, and $d_m = 2048$. We also extract Softmax-MM and GELU-MM from the entire layer, which are MMs of weights and the results from Softmax or GELU functions. The baseline is the conventional bit-serial CIM architecture. Our EBB-CIM architecture balances input bits across in-memory MACs. It reduces computation latency by 2.38× for Softmax-MM, 2.20× for GELU-MM, and 1.58× for the entire layer. The EB balancing causes only 5.1% power and 4.6% area overhead. Therefore, the energy savings are 2.19× for Softmax-MM, 2.02× for GELU-MM, and 1.45× for the entire layer.

VII. Experimental Results

This section first describes MulTCIM's measurement results from a 28-nm fabricated chip. We then provide the test

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Technology	TSMC 28nm CMOS	
Die Area	4.46mm×3.22mm	
Supply Voltage	0.6–1.0V	
Frequency	85–275MHz	
Buffer Size	192KB	
CIM Size	128KB	
Data Precision	INT8, INT16	
Work Mode	Pipeline/Parallel	
Power	29.83mW @0.6V, 85MHz	
	153.62mW @1.0V, 275MHz	
Peak Performance*1	INT8	3.55TOPS
	INT16	0.89TOPS
Area Efficiency*1	INT8	0.25TOPS/mm²
	INT16	0.06TOPS/mm²
Energy Efficiency*2*3	INT8	48.4-101.1TOPS/W
	INT16	12.1-60.3TOPS/W
Energy/Token*3	ViLBERT-base	2.24μJ/Token
	ViLBERT-large	7.72μJ/Token

*1: Measured at the highest performance point, 1.0V, 275MHz. Bit sparsity is exploited since it's an inherent data feature.
*2: Measured at the highest efficiency point, 0.7V, 160MHz.
*3: Low point: Only bit sparsity exploitation. High point: Peak efficiency evaluated on the ViLBERT-base model with attention-token-bit hybrid sparsity.

Fig. 16. MulTCIM's chip micrograph, VF scaling curves, and specifications.

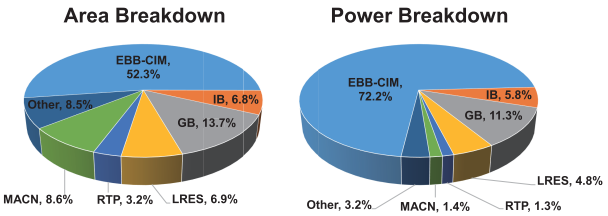


Fig. 17. MulTCIM's area and power breakdown. MACN: modal-adaptive CIM network. RTP: runtime token pruner. LRES: long reuse elimination scheduler. EBB-CIM: EB-balanced CIM. IB: input buffer. GB: global buffer.

platform settings for evaluation. We conduct comprehensive studies on the MulTCIM accelerator with typical multimodal Transformers (ViLBERT-base and ViLBERT-large [12]), including overall evaluation, layerwise analysis, and technique breakdown analysis. Finally, we compare MulTCIM with state-of-the-art Transformer accelerators and digital CIM accelerators.

A. Chip Micrograph and Summary

Fig. 16 presents MulTCIM's chip micrograph, voltage–frequency scaling curves, and specifications. MulTCIM is fabricated in 28-nm CMOS technology, with 4.46×3.22 mm^2 die area. The chip can work at 0.6–1.0 V supply, 85–275 MHz, with power consumption of 29.83–153.62 mW. MulTCIM is based on a full-digital SRAM-CIM architecture with INT8 and INT16 precision support. The peak performance reaches 3.55 TOPS at INT8 and 0.89 TOPS at INT16 under 1.0 V, 275 MHz. The highest energy efficiency is 101.1 TOPS/W at INT8 and 60.3 TOPS/W at INT16 under 0.7 V, 160 MHz, which is measured on the ViLBERT-base model with hybrid sparsity. Fig. 17 depicts MulTCIM's area and power breakdown. LRES consumes 6.9% in area and 4.8% in power. RTP consumes 3.2% in area and 1.3% in power. We separate the EBB-CIM macros from MACN to show the proportion of the inter-CIM network structure and MWA in MACN, which is 8.6% in area and 1.4% in power.

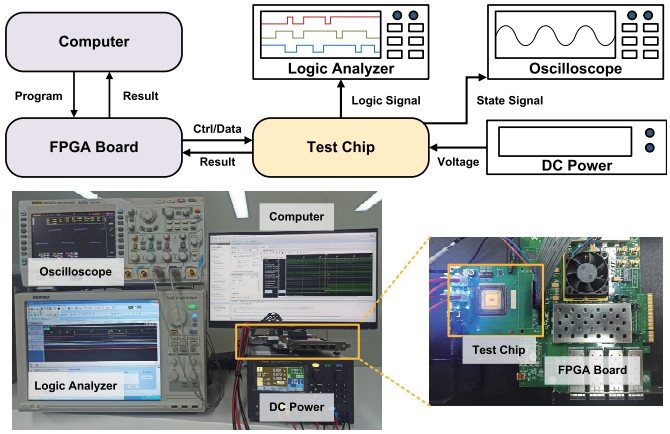


Fig. 18. Test platform for the fabricated MulTCIM accelerator.

TABLE I
OVERALL EVALUATION ON MULTIMODAL TRANSFORMERS

Model	ViLBERT-base	ViLBERT-large
Dataset	Visual Question Answering (VQA) v2.0	
Data Precision	INT16 for attention layers, INT8 for FC layers	
Accuracy (LA/SA)	70.23	71.48
Accuracy Loss*1 Attention, toke, bit	-0.29 (total) -0.04, -0.08, -0.17	-0.40 (total) -0.07, -0.10, -0.23
Execution Time*2	268ms	925ms
Speedup*2,3	9.47x (attention layers) 6.54x (entire model)	13.09x (attention layers) 7.58x (entire model)
Energy Saving*2,3	8.11x (attention layers) 5.61x (entire model)	11.22x (attention layers) 6.49x (entire model)
Energy/Token*2	2.24μJ/Token	7.72μJ/Token

*1: Compared with accuracy without sparsity at FP32. Metric: VQA score (test-dev).
*1: X is vision. Y is language. Token Count $N_X = 4096$, $N_Y = 1024$.
*2: Measured at 0.7V, 160MHz for high-efficiency evaluation.
*2: Off-chip memory is excluded to focus on chip-level evaluation.
*3: The baseline is similar to TranCIM, without optimizations for hybrid sparsity.

B. Test Platform Setup

Fig. 18 demonstrates the test platform for the fabricated MulTCIM accelerator. The system is composed of a MulTCIM test chip, a Xilinx VC709 FPGA board, dc power, an oscilloscope, a logic analyzer, and a host computer. The FPGA board's DDR3 memory stores the test dataset, multimodal Transformer model, and intermediate data produced during computation. Control signals and data are sent to MulTCIM via the FPGA mezzanine connector (FMC). The dc power supplies 0.6–1.0 V core voltage for the test chip. MulTCIM chip's computing results are collected by the FPGA and then sent to the host computer for further analysis.

C. Evaluation on Multimodal Transformers

1) Overall Evaluation: Table I presents MulTCIM's overall evaluation on two typical multimodal Transformer models, ViLBERT-base and ViLBERT-large [12], with the visual question answering (VQA) v2.0 dataset. The attention layers use INT16 precision, and the FC layers use INT8 precision. The sparsity settings for the two benchmarks are as follows: Attention sparsity = 70%. Modal X's token sparsity = 30%. Modal Y's token sparsity = 50%. Softmax-MSB bit sparsity = 58.03%. GELU-MSB bit sparsity = 54.60%. On ViLBERT-base, attention, token, and bit sparsity cause accuracy loss of -0.04, -0.08, and -0.17. On ViLBERT-large, attention,

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TABLE II
LAYERWISE ENERGY ANALYSIS ON MULTIMODAL TRANSFORMERS

Layer Type		ViLBERT-base		ViLBERT-large	
Energy (μJ)		Baseline	Ours	Baseline	Ours
Modal X Encoders	QK^T-MM	2672.93	299.29	4009.39	448.93
	$A'V$-MM	2433.05	103.95	3649.57	93.56
	FC Layer	599.69	290.08	1349.31	652.68
	One Stack	5705.67	693.32	9008.27	1195.17
	Total	22822.67	2773.28	72066.19	9561.33
Modal Y Encoders	QK^T-MM	385.52	83.08	618.97	157.06
	$A'V$-MM	295.56	17.31	446.56	69.77
	FC Layer	337.33	120.78	599.69	214.72
	One Stack	1018.41	221.17	1665.22	441.56
	Total	4073.64	884.70	13321.79	3532.47
Cross-Modal Encoders	QK^T-MM	2030.40	443.34	4615.24	620.68
	$A'V$-MM	1700.56	121.27	4135.49	163.33
	FC Layer	937.02	410.86	1949.01	867.40
	One Stack	4667.98	975.47	10699.73	1651.41
	Total	37343.86	7803.76	171195.75	26422.54
Entire Model	Total	64240.17	11461.73	256583.73	39516.35

token, and bit sparsity cause accuracy loss of −0.07, −0.10, and −0.23. The bit sparsity's accuracy loss is caused by INT8/16 quantization, while the proposed effective-bit balancing has no impact on accuracy. The long reuse elimination technique has no impact on accuracy as it just reshapes the attention matrix without change the values. Similarly, symmetric modal overlapping and effective-bit balancing also have no accuracy loss as they just change the computation order.

The performance and energy results in Table I are obtained at 0.7 V, 160 MHz for high-efficiency measurement. Off-chip memory is excluded to focus on chip-level evaluation. The baseline is similar to the architecture of TranCIM [20], which has no optimization techniques for hybrid sparsity. On ViLBERT-base, our work achieves 9.47× speedup and 8.11× energy saving for attention layers, contributing to 6.54× speedup and 5.61× for the entire model. On ViLBERT-large, our work achieves 13.09× speedup and 11.22× energy saving for attention layers, contributing to 7.58× speedup and 6.49× for the entire model. Due to high weight reuse and pipeline overlapping, CIM updating consumes 1.09% latency, 0.01% energy on ViLBERT-base, and 1.03% latency, and 0.006% energy on ViLBERT-large. We will take a closer look at the CIM behaviors in the parallel and pipeline modes. In the parallel mode and StageS of the pipeline mode, weights are statically loaded into CIM and kept reused during computation. The CIM updating overhead is amortized by the higher computation latency and energy. In StageD of the pipeline mode, the CIM updating latency is overlapped. LRES ensures that weights are written into StageD's CIM only once, thus reducing updating energy. We also measure the metric "Energy/Token," which equals the average energy consumption of processing one input token in the Transformer models. MulTCIM achieves only 2.24 μJ/Token on ViLBERT-base, and 7.72 μJ/Token on ViLBERT-large.

2) Layerwise Analysis: Table II presents layerwise analysis on ViLBERT-base and ViLBERT-large inference at 0.7 V, 160 MHz. Since multimodal Transformers are constructed by single-modal and cross-modal encoders (see Fig. 1), we separate the latency and energy into encoders for Modal X, Modal Y, and cross-modal, where X stands for vision and Y stands for language. Each encoder is a stacking structure with multiple attention layers and FC layers, so we also report evaluation results on one stack and provide breakdown analysis on attention layers (QK^T-MM and $A'V$-MM) and FC layers. Table II compares the energy consumption of MulTCIM with the baseline that has no optimizations for hybrid sparsity. Let us take a closer look at the ViLBERT-base model. In the Modal X encoders, attention layers' energy is reduced by 12.66×. In the Modal Y encoders, attention layers' energy is reduced by 6.78×. MulTCIM works better for the Modal X encoders that have larger token count. This is because the baseline's full attention computation has a quadratic relation with token count. MulTCIM's attention sparsity significantly reduces the computation complexity, and token sparsity further removes insignificant tokens. The conclusion is similar for the encoders in the ViLBERT-large model.

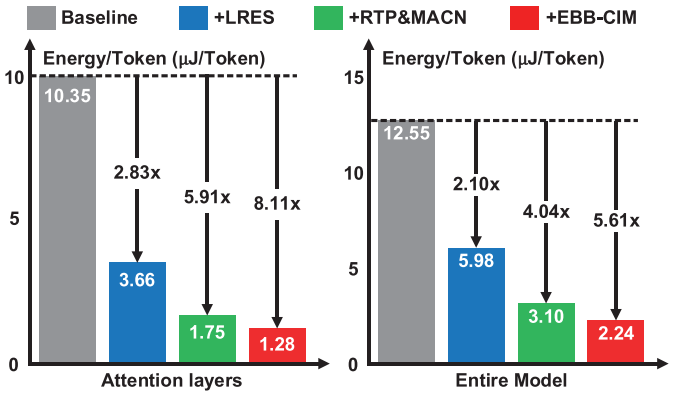


Fig. 19. Technique breakdown analysis on ViLBERT-base.

3) Technique Breakdown Analysis: Fig. 19 demonstrates a technique breakdown analysis on the ViLBERT-base benchmark, to clearly illustrate how the energy consumption is saved with MulTCIM's features. Since our techniques focus more on the sparsity opportunities of attention layers, we also provide breakdown analysis on the attention layers besides the entire model. The baseline can support the pipeline/parallel reconfigurable modes, but has no optimization techniques for hybrid sparsity (gray bar). On the attention layers, we apply attention sparsity with LRES (blue bar), achieving energy saving of 2.83×. We then utilize RTP and MACN to prune insignificant tokens, further reducing the attention layer's energy consumption to 5.91×. Finally, EBB-CIM introduces bit sparsity exploitation to attention layers, so the energy saving reaches 8.11× over the baseline. On the entire model, although FC layers cannot obtain benefits from attention sparsity, we can still observe significant energy saving provided by the three levels of sparsity (2.10×, 1.93×, and 1.38×, respectively).

4) Sparsity Scaling Analysis: Fig. 20(a) sweeps attention sparsity on the ViLBERT-base model without token pruning (token sparsity = 0). The dashed lines indicate the latency and energy if LRES is disabled and attention sparsity is 0. The latency and energy become both lower than the baseline if attention sparsity ≥30%. This is because lower sparsity makes the attention pattern similar to global attention and there is no much space to reshape the attention pattern. Then, we set

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TABLE III
COMPARISON WITH STATE-OF-THE-ART TRANSFORMER ACCELERATORS AND DIGITAL CIM ACCELERATORS

	ISSCC'22-29.2 [22]	ISSCC'22-11.6 [6]	ISSCC'22-29.3 [20]	This Work
MAC Implementation	Digital MAC	Digital CIM	Digital CIM	Digital CIM
Data Precision	INT12	INT4/8	INT8/16	✓ INT8/16
Transformer Sparsity Level	Attention	N/A	Attention	✓ Attention/Token/Bit
Work Mode	Parallel	Parallel	Pipeline/Parallel	✓ Pipeline/Parallel
Technology	28nm	5nm	28nm	28nm
Supply Voltage (V)	0.56∼1.1	0.5∼0.9	0.6∼1.0	0.6∼1.0
Frequency (MHz)	50∼510	360∼1440	80∼240	85∼275
Die Area (mm^2)	6.82	0.0133 (macro)	6.83	14.36
Power (mW)	12.06∼272.8	11.61	27.04∼118.21	28.83∼152.75
Peak Performance (TOPS)^{*1}	4.07 (INT12)	0.74 (INT8), 0.18 (INT16)^{*2}	1.48 (INT8), 0.37 (INT16)	3.55 (INT8), 0.89 (INT16)^{*4}
Energy Efficiency (TOPS/W)^{*1}	27.56 (INT12)	44.76 (INT8), 11.19 (INT16)^{*2,3}	20.5 (INT8), 5.1 (INT16)	101.1 (INT8), 60.3 (INT16)^{*5,6}
Norm. Area Efficiency (TOPS/mm^2)^{*1}	0.13 (INT8)^{*2}	0.19 (INT8)	0.22 (INT8)	0.25 (INT8)^{*4}
ViLBERT-base Energy (μJ/Token)	13.23 (5.91x)	5.60 (2.50x)^{*2,3}	12.55 (5.61x)	2.24 (1x)^{*5,6}
and Energy Efficiency (TOPS/W)^{*1}	14.51	34.30 ^{*2,3}	15.28	85.75^{*5,6}

^{*1}: One operation (OP) represents one multiplication or one addition.
^{*2}: [6]'s INT16 results are projected from the reported INT4/8 numbers. [22]'s INT8 results are projected from the reported INT12 numbers.
^{*3}: On-chip buffers are included for fair comparison. Buffer capacity is 192KB, same as MulTCIM.
^{*4}: Measured at the highest performance point, 1.0V, 275MHz. Bit sparsity is exploited since it's an inherent data feature.
^{*5}: Measured at the highest efficiency point, 0.7V, 160MHz, on the ViLBERT-base model.
^{*6}: Attention sparsity = 70%. Modal X's token sparsity = 30%. Modal Y's token sparsity = 50%. Softmax-MSB bit sparsity = 58.03% and GELU-MSB bit sparsity = 54.60%.

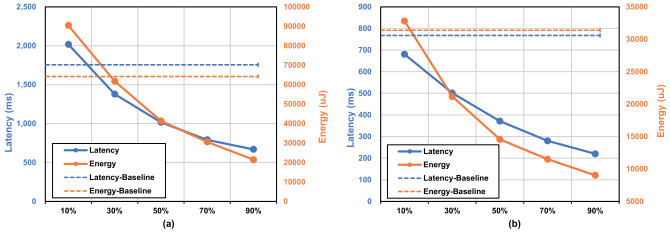


Fig. 20. Sparsity scaling analysis on ViLBERT-base. (a) Attention sparsity scaling. (b) Token sparsity scaling.

the attention sparsity as 70% and sweep token sparsity [see Fig. 20(b)]. The token sparsity just needs to be higher than 10% to make latency and energy lower than the baseline. This is because RTP only consumes 1.3% in power and pruning tokens directly reduces the computation amount. We do not sweep bit sparsity because it is not a tunable parameter. The bit sparsity exploited by MulTCIM is an inherent data feature determined by activation functions.

D. Comparison With State-of-the-Art Works

Table III shows a detailed comparison with state-of-the-art Transformer accelerators and digital CIM accelerators. Since Transformer models usually require precision higher than INT8 to maintain accuracy, we only include digital architecture and digital CIM baselines for comparison. ISSCC'22-29.2 [22] is a digital Transformer accelerator with approximation computing techniques for weak attention. ISSCC'22-11.6 [6] is a digital CIM macro designed by TSMC. ISSCC'22-29.3 [20] is TranCIM, a digital CIM-based Transformer accelerator with pipeline/parallel reconfigurable modes. MulTCIM is designed on the promising digital CIM-based architecture with full exploitation of attention-token-bit hybrid sparsity.

We normalize the four works' area efficiency to the same technology node, voltage, frequency, and INT8 precision. Thus, the metric TOPS/mm^2 can fairly reflect the performance/area achieved by different architectures. Bit sparsity is considered in the measurement since it is an inherent data feature. Although MulTCIM has larger silicon area, the digital CIM architecture with bit sparsity exploitation helps achieve much higher area efficiency, 1.12×–1.96× higher than previous work. The two last row of Table III evaluates the energy consumption and energy efficiency of the four works on the ViLBERT-base model. MulTCIM consumes only 2.24 μJ/Token and 85.75 TOPS/W for the benchmark, achieving 5.91×, 2.50×, and 5.61× energy reduction over [6], [20], and [22], respectively. MulTCIM's advantages over [22] come from the MAC implementation and sparsity exploitation. Digital CIM provides a more efficient MM architecture, and the additional support for token-bit sparsity enhances the efficiency. Fujiwara et al. [6] only reported evaluation results on the CIM macro, so we add 192-kB on-chip buffers (same capacity as MulTCIM) to it for chip-level comparison. Although Fujiwara et al. [6] have more advanced fabrication technology, MulTCIM's hybrid sparsity and flexible inter-CIM network help achieve even higher energy efficiency. Tu et al. [20] is a state-of-the-art Transformer accelerator based on digital CIM. Although it can support attention sparsity of vanilla Transformers, the sparsity should be in a static and regular pattern. In multimodal Transformers, attention sparsity is usually dynamic and irregular, which degrades the efficiency of Tu et al.'s [20] architecture. In MulTCIM, the irregular attention sparsity can be dynamically reorganized by LRES for higher CIM utilization. Besides, MulTCIM also introduces attention and bit sparsity that is not exploited in [20]. The proposed hybrid sparsity techniques can be applied to both multimodal and vanilla Transformers.

VIII. CONCLUSION

This article proposes MulTCIM, a digital CIM-based accelerator for multimodal Transformers. To improve energy efficiency, MulTCIM exploits attention-token-bit hybrid sparsity, which is not fully investigated in previous Transformer accelerators. The key idea of MulTCIM is handling the irregular dataflow caused by hybrid sparsity, so the reduced computation can be better converted to efficiency improvement.

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MulTCIM has three features for utilizing the hybrid sparsity: LRES dynamically reshapes the attention pattern to improve CIM utilization. RTP removes insignificant tokens and MACN exploits symmetric modal overlapping to reduce CIM idleness. EBB-CIM balances input bits across in-memory MACs to reduce computation time. The fabricated MulTCIM consumes only 2.24 μJ/Token for the ViLBERT-base model, achieving 2.50×–5.91× lower energy than previous Transformer accelerators and digital CIM accelerators.

MulTCIM's design insight is that inter- and intra-CIM architecture optimizations can help construct flexible dataflows to broaden the application scope of digital CIM. Most current digital CIM research focuses on macro-level circuit design, but CIM macros are just basic computing units in a CIM-based AI accelerator. As AI models are getting larger and more complex, it is quite essential to optimize both inter- and intra-CIM architecture to achieve high efficiency at the accelerator level. Future work can consider how to scale out MulTCIM for the emerging large language models (LLMs) or apply the design insight to other application domains.

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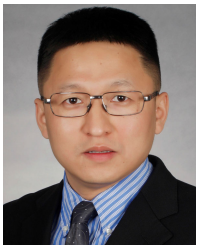
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